

9_Operators_part2

September 15, 2026

```
[17]: a = 5  
      bin(5)
```

```
[17]: '0b101'
```

```
[19]: print(int('0b1101',base=2))
```

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```
[ ]: # Binary Gates  
      # AND, OR and XOR gates  
      # A | B | AND | OR | XOR (Inequality Detector)  
      # 0 | 0 | 0 | 0 | 0  
      # 0 | 1 | 0 | 1 | 1  
      # 1 | 0 | 0 | 1 | 1  
      # 1 | 1 | 1 | 1 | 0
```

```
[31]: a=5 #0101 is binary representation  
      b=3 #0011 is binary representation  
      print(" a & b results is ", a & b)  
      print(" a & b binary representation is ", bin(a & b))  
      print(" a & b value decimal representation is ",int(bin(a&b),base=2))  
      print(" a|b results is ", a | b )  
      print(" a|b binary representation is ",bin(a | b))  
      print(" a|b value decimal representation is ",int(bin(a|b), base=2))
```

```
a & b results is 1  
a & b binary representation is 0b1  
a & b value decimal representation is 1  
a|b results is 7  
a|b binary representation is 0b111  
a|b value decimal representation is 7
```

```
[38]: # Negation (bitwise Not) represented by symbol/character "~"  
      # Decimal 5 --> represented by 0101 in binary + 1 (0001)  
      # Result is 0110 which 6 but negation will invert the sign of the give input  
      # so +5 bitwise not results is -6  
      # Generic formulae is ~n= -(n+1)
```

```
[40]: a = 5
      print(~a)
```

-6

```
[42]: for i in range(-10,10,1):    # Iterating over -10 to +9 to apply bitwise NOT on
      ↪each number to see the results
      print("negation of {} is {}".format(i,~i)) #Ignore the syntax as of now as
      ↪this is for understanding bitwise NOT results for each given number
```

```
negation of -10 is 9
negation of -9 is 8
negation of -8 is 7
negation of -7 is 6
negation of -6 is 5
negation of -5 is 4
negation of -4 is 3
negation of -3 is 2
negation of -2 is 1
negation of -1 is 0
negation of 0 is -1
negation of 1 is -2
negation of 2 is -3
negation of 3 is -4
negation of 4 is -5
negation of 5 is -6
negation of 6 is -7
negation of 7 is -8
negation of 8 is -9
negation of 9 is -10
```

```
[48]: # Left Shift (bitwise Left << n), here n represents how many times the left
      ↪shift should happen
      a=5          # 5 (decimal) --> 0101 (binary) if we move the values to left
      ↪then 1010 (binary) = 10 (decimal)
      print(a << 1)
```

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```
[47]: # Right Shift (bitwise right >> n), In general one left shift will make value
      ↪double
      # Right shift will make the value into half
      b = 6         # 6 (decimal) --> 0110 (binary) if we move the values to the
      ↪right then 0011 (binary) = 3 (decimal)
      print(b >> 1)
```

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